

This spreadsheet contains various test cases for lookup functions: VLOOKUP, HLOOKUP, LOOKUP.
Name of the main junit test class which uses this spreadsheet:
`org.apache.poi.ss.formula.functions.TestLookupFunctionsFromSpreadsheet`
(The content of cell 'Read Me'!A3 is confirmed by the test)

Every sheet besides this first one contains formula evaluation test cases in a standardised format.
On row 4 of each sheet, in columns B,C,D there are the column headings "Formula", "Expected Result", "Actual Result".
The test iterates down from row 5 onward until the text "<end>" is found in column A.
Rows with "<skip>" in column A are ignored (useful for currently unsupported behaviour).
Otherwise column A can be used for commenting the group of rows below.

If the evaluated result of column B does not match the expected result in column C, the junit test will report a failure. Test failures get annotated with the section and row number.
Besides the first 4 columns, row 5 onwards, the junit test does not inspect any other cells in the sheet. The other cells can be used freely to set up test data.
Care should be taken to not only make the values in column C match those in column B, but also to make sure column C contains only simple literal values. This can be achieved easily by 'pasting special' from column B to C, selecting the 'as values' option.

ed Result" and "Comment"

Sheet Comment: Simple tests of VLOOKUP using basic argument valu

Formula Expected Result

r3 r3
r1 r1

#N/A #N/A

r3 r3
r2 r2
r6 r6

#N/A #N/A
#N/A #N/A
#N/A #N/A

r9 r9
rZero rZero

0 0

bad col index param

#REF! #REF!
#REF! #REF!
#REF! #REF!
#REF! #REF!
#REF! #REF!
#REF! #REF!
#N/A #N/A
#N/A #N/A
#VALUE! #VALUE!
#REF! #REF!

string lookup col

#N/A #N/A

32 32

Simple values wilcard something1

something1 something1
something1 something1
something3 something3
something3 something3
something3 something3
something2 something2
something7 something7
#N/A #N/A
#N/A #N/A
something8 something8

something9 something9

something10	something10
something11	something11
something13	something13
something14	something14
something15	something15

something4	something4
#N/A	#N/A
something4	something4
something4	something4
something4	something4
something4	something4
#N/A	#N/A
something5	something5
something5	something5
#N/A	#N/A
#N/A	#N/A
#N/A	#N/A
#N/A	#N/A
#N/A	#N/A
#N/A	#N/A
#N/A	#N/A

Complex values wildcard

something1	something1
something1	something1
something3	something3
something3	something3
something3	something3
something2	something2

something7
#N/A
#N/A
something8

something9

something10

something11

something13

something14

something15

something7
#N/A
#N/A
something8

something9

something10

something11

something13

something14

something15

something4
#N/A
something4
something4
something4
something4
#N/A
something5
something5
#N/A

something4
#N/A
something4
something4
something4
something4
#N/A
something5
something5
#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

<end>

es

Comment	lookup_value	col_index_num
first occurence of 30	45	2
	30	2
	55	2
last occurence of 30	45	2
	30	2
	55	2
blank lookup val empty cell result empty text result	2	2
	-5	2
	90	2
	90	2
		2
	75	2
	80	2
	0	#VALUE!
	0	#DIV/0!
	0	#N/A
#N/A	0	#REF!
	0	#NUM!
	0 abc	
		#DIV/0
	0	#DIV/0
	0	
	0	6
blank lookup does not match empty string		2
empty string matches empty string		2
only find an exact match.	Bla*	2
only find an exact match with double wildcard	*Bla*	2
	Bla?45	2
	Bla?4?	2
	Bla?4*	2
having BLA + 5 any letters	BLA?????	2
	?????BLA	2
	CanNotBefound*	2
	CanNotBefou?d	2
	POI for	2
literal characters that have special meaning in regex	*bugs~?	2

literal characters that have special meaning in regex	*now [~] ? [~] ?	2
literal characters that have special meaning in regex	3 [~] *2=6	2
literal characters that have special meaning in regex	\$not*	2
literal characters that have special meaning in regex	*[2*	2
literal characters that have special meaning in regex	* [^]	2

Bla*	2
Bla	2
Bla?45	2
Bla?4?	2
Bla?4*	2
BLA?????	2
????BLA	2
CanNotBefound*	2
CanNotBefou?d	2
POI for	2

literal characters that have special meaning in regex	*bugs [~] ?	2
literal characters that have special meaning in regex	*now [~] ? [~] ?	2
literal characters that have special meaning in regex	3 [~] *2=6	2
literal characters that have special meaning in regex	\$not*	2
literal characters that have special meaning in regex	*[2*	2
literal characters that have special meaning in regex	* [^]	2

value column & *	Bla	2
value * & column & *	Bla	2
value &"?45"	Bla	2
value &"?4?"	Bla	2
value &"?4*"	Bla	2
value &"?????"	BLA	2

"???"& value	BLA	2
	CanNotBefound	2
	CanNotBefou	2
"*&value&*"	POI for	2
literal characters that have special meaning in regex	bugs	2
literal characters that have special meaning in regex	now	2
literal characters that have special meaning in regex	2=6	2
literal characters that have special meaning in regex	\$not	2
literal characters that have special meaning in regex	[2	2
literal characters that have special meaning in regex	*^	2

	Bla	2
	Bla	2
	Bla	2
	Bla	2
	Bla	2
	BLA	2
	BLA	2
	CanNotBefound	2
	CanNotBefou	2
	POI for	2
literal characters that have special meaning in regex	bugs	2
literal characters that have special meaning in regex	now	2
literal characters that have special meaning in regex	3	2
literal characters that have special meaning in regex	\$not	2
literal characters that have special meaning in regex	[2	2
literal characters that have special meaning in regex	^	2

FALSE

3*2=6

FALSE

(Apache)(POI)

FALSE

\$not beginning

FALSE

simple vector [2,3,4]

FALSE

not at the end ^

If TRUE or omitted, an approximate match is returned. In other words, if an exact match is not found, the next largest value that is less than lookup_value is returned.

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

If FALSE, HLOOKUP will find an exact match. If one is not found, the error value #N/A is returned.

FALSE

FALSE

FALSE

FALSE

FALSE

FALSE

FALSE
FALSE
FALSE
FALSE

FALSE

FALSE

FALSE

FALSE

FALSE

FALSE

If TRUE or omitted, an approximate match is returned. In other words, if an exact match is not found, the next largest value that is less than lookup_value is returned.

TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE
TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

TRUE

r-2	
rZero	
r0	24
r1	25
r2	26
r3	27
r4	28
r5	29
r6	30
	31
	32
r9	33

col_index_num

something1
something2
something3
something4
something5
something6
something7
something8
something9

something10

something11

something12

something13

something14

something15

Sheet Comment: Tests of VLOOKUP using some unusual argument values

Formula	Expected Result	Comment	lookup_value
Odd values for col_index_num			
	47	47 key column	47
r5	r5		47
s5	s5	something besides 2	47
#REF!	#REF!	column 4 out of range	47
r5	r5	scientific notation	47
r5	r5	scientific notation	47
r5	r5		47
#VALUE!	#VALUE!		47
#REF!	#REF!		47
<skip> r5	r5	text scientific notation not supported	47
	47	47	47
#VALUE!	#VALUE!		47
#REF!	#REF!		47
#REF!	#REF!		47
#VALUE!	#VALUE!	blank	47
Odd values for range_lookup			
r2	r2	no range_lookup arg	30
r1	r1	blank	30
r1	r1	text false	30
r2	r2	text true	30
#VALUE!	#VALUE!	text 1	30
r2	r2		30
r1	r1		30
r2	r2		30
r2	r2		30
#N/A	#N/A	not exact, range lookup	35
r2	r2	not exact, no range lookup	35
r2	r2	not exact, no range argument	35
#N/A	#N/A	not exact, empty range argument	35
r2	r2	exact, no range argument	30
r1	r1	exact, empty range argument	30
Out of range lookup_value:			
#N/A	#N/A		2
#N/A	#N/A		2
#N/A	#N/A		90
r9	r9		90
<end>			

Lookup Table

col_index_num	range_lookup		
		5 r0	s0
		30 r1	s1
		30 r2	s2
		45 r3	s3
		46 r4	s4
		47 r5	s5
		50 r6	s6
		75 r7	s7
		80 r8	s8
		81 r9	s9
	1	FALSE	
	2	FALSE	
	3	FALSE	
	4	FALSE	
	2. 99	FALSE	
	2. 99E+00	FALSE	
2		FALSE	
0. 9		FALSE	
5. 1		FALSE	
29. 9e-1		FALSE	
TRUE		FALSE	
FALSE		FALSE	
TRUE		FALSE	
FALSE		FALSE	
		FALSE	

2 <arg not passed>
2
2 false
2 TRUE
2 1
2 1
2 0
2 -0. 01
2 0. 01
2 FALSE
2 TRUE
2 <arg not passed>
2 <arg empty>
2 <arg not passed>
2 <arg empty>

2 FALSE
2 TRUE
2 FALSE
2 TRUE

Sheet Comment: Tests of unspecified behaviour of VLOOKUP when the table ar
For all these tests range_lookup=TRUE and col_num=2

Formula	Expected Result	Comment	lookup_value
<skip>r9	r9	tab 1	33
<skip>r3	r3	tab 2	33
<skip>r5	r5	tab 3	33
<skip>r3	r3	tab 4	33
<skip>r15	r15	tab 5	33
<skip>r10	r10	tab 5	10
r07	r07	tab 6	33

<end>

ray is not sorted

Lookup Table 1

33 r0
33 r1
33 r2
100 r3
33 r4
33 r5
33 r6
33 r7
33 r8
33 r9

Lookup Table 2

33 r0
33 r1
33 r2
33 r3
100 r4
33 r5
33 r6
33 r7
33 r8
33 r9

Lookup Table 3

33 r0
33 r1
33 r2
33 r3
33 r4
33 r5
5 r6
33 r7
100 r8
33 r9

Lookup Table 4

32 r0
32 r1
32 r2
31 r3
100 r4
32 r5
32 r6
32 r7
32 r8
32 r9

Lookup Table 5

33 r00
33 r01
33 r02
33 r03
33 r04
33 r05
33 r06
33 r07
33 r08
33 r09
8 r10
33 r11
33 r12
6 r13
33 r14
33 r15
4 r16
33 r17
33 r18
33 r19
33 r20

Lookup Table 6

33 r00
33 r01
33 r02
33 r03
10 r04
33 r05
33 r06
33 r07
5 r08
33 r09
100 r10
33 r11
33 r12
33 r13
33 r14
33 r15
33 r16
33 r17
33 r18
33 r19
22 r20

Sheet Comment:

Formula

2

r3
r1
#N/A

r3
r2
r6

#N/A
#N/A
#N/A

r9

bad row arg

#REF!
#VALUE!
#REF!
#VALUE!

Wildcard values

something1
something2
something6
something3
something7
something2
something5
#N/A
#N/A
#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

something4
something4
 #N/A
 #N/A
something6
something4
something4
something5
something5
#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

Complex Wildcards values

something1
something2
something6
something3
something7
something2
something5
#N/A
#N/A
#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

#N/A

<end>

Simple tests of HLOOKUP using basic argument values

Expected Result		Comment	lookup_value
		2	
r3			45
r1		first occurence of 30	30
	#N/A		55
r3			45
r2		last occurence of 30	30
r6			55
	#N/A		2
	#N/A		2
	#N/A		90
r9			90
	#REF!		45
	#VALUE!		45
	#REF!	col index too high	45
	#VALUE!		45
	something1		BLA*
	something2		BLA2*
	something6		*IER3*
	something3		*IER1*
	something7		SOME???
	something2		BLA?3567
	something5		BLA*34
	#N/A		CanNotBefound*
	#N/A		CanNotBefou?d
	#N/A		*POI for*
		literal characters that have special meaning in regex	*bugs~?
	#N/A	literal characters that have special meaning in regex	*now~?~?
		literal characters that have special meaning in regex	3~*2=6

	literal characters that have special meaning in regex	\$not*
#N/A	literal characters that have special meaning in regex	
	literal characters that have special meaning in regex	*[2*
#N/A	literal characters that have special meaning in regex	*^
something4		BLA*
something4		BLA2*
#N/A		*IER3*
#N/A		*IER1*
something6		SOME???
something4		BLA?3567
something4		BLA*34
something5		CanNotBefound*
something5		CanNotBefou?d
#N/A		*POI for*
	literal characters that have special meaning in regex	
#N/A	literal characters that have special meaning in regex	*bugs~?
	literal characters that have special meaning in regex	
#N/A	literal characters that have special meaning in regex	*now~?~?
	literal characters that have special meaning in regex	
#N/A	literal characters that have special meaning in regex	3~*2=6
	literal characters that have special meaning in regex	
#N/A	literal characters that have special meaning in regex	\$not*
	literal characters that have special meaning in regex	
#N/A	literal characters that have special meaning in regex	*[2*

#N/A	literal characters that have special meaning in regex	*^	
something1		BLA	
something2		BLA2	
something6		IER3	
something3		IER1	
something7		SOME	
something2		BLA	
something5		BLA	
#N/A		CanNotBefound	
#N/A		CanNotBefou	
#N/A		POI for	
	literal characters that have special meaning in regex		
#N/A	literal characters that have special meaning in regex	bugs	
	literal characters that have special meaning in regex		
#N/A	literal characters that have special meaning in regex	now	
	literal characters that have special meaning in regex		
#N/A	literal characters that have special meaning in regex		3
	literal characters that have special meaning in regex		
#N/A	literal characters that have special meaning in regex	\$not	
	literal characters that have special meaning in regex		
#N/A	literal characters that have special meaning in regex	[2	
	literal characters that have special meaning in regex		
#N/A	literal characters that have special meaning in regex	^	

		Lookup Table			
col_index_num	range_lookup		5	30	30
		r0	r1	r2	
	2	FALSE			
	2	FALSE			
	2	FALSE			
	2	TRUE			
	2	TRUE			
	2	TRUE			
	2	FALSE			
	2	TRUE			
	2	FALSE			
	2	TRUE			
#DIV/0		TRUE			
		TRUE			
	5	TRUE			
	-1	TRUE			

If FALSE, HLOOKUP will find an exact match. If one is not found, the error value #N/A is returned.

2	FALSE
2	FALSE
2	FALSE
2	FALSE
2	FALSE
2	FALSE
2	FALSE
2	FALSE
2	FALSE
2	FALSE

2	FALSE
---	-------

2	FALSE
---	-------

2	FALSE
---	-------

Lookup Table		
BLA16	BLA23567	MEIER1234
something1	something2	something3

2 FALSE

2 FALSE

2 FALSE

If TRUE or omitted, an approximate match is returned. In other words, if an exact match is not found, the next largest value that is less than lookup_value is returned.

2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE
2 TRUE

2 TRUE

2 TRUE

2 TRUE

2 TRUE

2 TRUE

2 TRUE
If FALSE, HLOOKUP will
find an exact match.
If one is not found, the
error value #N/A is
returned.

2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE
2 FALSE

2 FALSE

2 FALSE

2 FALSE

2 FALSE

2 FALSE

2 FALSE

	45	46	47	50	75	80
r3	r4	r5	r6	r7	r8	

BLA	BLATST234	MEIER3456	SOMEBLA	Apache POI for exce	Got bugs?
something4	something5	something6	something7	something8	something9

r9

Oh...is this a bug now?? 3*2=6 (Apache)(POI) \$not beginning
something10 something11 something12 something13


```
simple vector [2,3,4] not at the end ^  
something14      something15
```

Sheet Comment: Simple tests of LOOKUP using basic argument values

Formula	Expected Result	Comment	lookup_value
both vectors columns			
r3	r3		45
r2	r2	last occurence of 30	30
r6	r6		55
#N/A	#N/A		2
r9	r9		90
both vectors rows			
r3	r3		45
r2	r2	last occurence of 30	30
r6	r6		55
#N/A	#N/A		2
r9	r9		90
lookup column result row			
r3	r3		45
r2	r2	last occurence of 30	30
r6	r6		55
#N/A	#N/A		2
r9	r9		90
lookup row result column			
r3	r3		45
r2	r2	last occurence of 30	30
r6	r6		55
#N/A	#N/A		2
r9	r9		90
r8	r8		80

<end>

Lookup Table

5 r0
30 r1
30 r2
45 r3
46 r4
47 r5
50 r6
75 r7
80 r8
81 r9

Lookup Table

5 30 30 45 46 47 50 75 80 81
r0 r1 r2 r3 r4 r5 r6 r7 r8 r9